

Defⁿ:

Rank of Matrix

Let $A_{m \times n}$ be a matrix of order $m \times n$ then the positive integer 'r' is called Rank of matrix A if

- i) there exists a non-zero minor of order 'r', and
- ii) all the minors of order $r+1$ and more are zero.

Defⁿ:

The nos. of non-zero rows (columns) in row-echelon form (or column-echelon form) is called Rank of matrix and is denoted by

$\rho(A)$.

If A is of order $m \times n$ then

$$\rho(A) \leq \min(m, n)$$

Row-Echelon Form: (Upper Triangular Matrix)

$$A = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \left. \begin{array}{l} \text{Non-zero rows} \\ \text{Zero-row} \end{array} \right\}$$

Column Echelon Form: (Lower Triangular Matrix) $\therefore \rho(A) = 2$

$$A = \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix}$$

Find the rank of

$$A = \begin{bmatrix} 7 & 4 & 2 & -8 \\ 5 & 3 & 12 & -15 \\ 6 & 4 & 14 & 2 \end{bmatrix}_{3 \times 4}$$

Solⁿ: Let $A = \begin{bmatrix} 7 & 4 & 2 & -8 \\ 5 & 3 & 12 & -15 \\ 6 & 4 & 14 & 2 \end{bmatrix}$

$$\therefore R_1 - R_3 \sim \begin{bmatrix} 1 & 0 & -12 & -10 \\ 5 & 3 & 12 & -15 \\ 6 & 4 & 14 & 2 \end{bmatrix}$$

$$R_2 - 5R_1, R_3 - 6R_1$$

$$\sim \begin{bmatrix} 1 & 0 & -12 & -10 \\ 0 & 3 & 72 & 35 \\ 0 & 4 & 86 & 62 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$\sim \begin{bmatrix} 1 & 0 & -12 & -10 \\ 0 & 4 & 86 & 62 \\ 0 & 3 & 72 & 35 \end{bmatrix}$$

$$R_2 - R_3$$

$$\sim \begin{bmatrix} 1 & 0 & -12 & -10 \\ 0 & 1 & 14 & 27 \\ 0 & 3 & 72 & 35 \end{bmatrix}$$

$$R_3 - 3R_2 \sim \begin{bmatrix} 1 & 0 & -12 & -10 \\ 0 & 1 & 14 & 27 \\ 0 & 0 & 30 & -46 \end{bmatrix}$$

$$\therefore \rho(A) = 3$$